

End-to-End E-commerce Intelligence System Building a Customer 360 Analytics Framework

1. Business Problem Statement

Modern e-commerce platforms generate large amounts of data across multiple independent systems such as customer management, orders, products, sellers, reviews, and payments. This project integrates multiple relational datasets into a unified analytical framework to generate actionable business insights.

2. Dataset Description

The project uses 9 relational datasets including customers, orders, order_items, payments, reviews, products, sellers, geolocation, and category translation datasets.

3. Methodology

The project workflow includes data loading, preprocessing, data integration, feature engineering, exploratory data analysis, visualization, and business insight generation.

4. Data Cleaning and Preprocessing

Performed missing value handling, duplicate removal, datetime conversion, data validation, and column standardization using Pandas.

5. Data Integration

Merged multiple relational tables to create a unified master dataset called master_df using customer_id, order_id, product_id, and seller_id as primary keys.

6. Feature Engineering

Created business features such as total_order_value, delivery_time_days, items_per_order, customer_purchase_frequency, customer_lifetime_value, and average_order_value.

7. Exploratory Data Analysis (EDA)

Analyzed customer behavior, revenue trends, product performance, seller contribution, and customer satisfaction patterns.

8. Data Visualization

Created time series plots, bar charts, histograms, box plots, and heatmaps using Matplotlib and Seaborn.

9. Key Findings

Repeat customers contribute significantly to revenue. Faster deliveries improve customer satisfaction. Some product categories and sellers dominate revenue generation.

10. Business Insights and Recommendations

Improve delivery efficiency, strengthen customer retention strategies, support high-performing sellers, and optimize operational performance.

11. Conclusion

This project demonstrates a complete end-to-end e-commerce analytics workflow and provides a strong foundation for future machine learning applications.

Dataset Overview

Dataset	Description
customers.csv	Customer information
orders.csv	Order lifecycle details
order_items.csv	Product-level order information
payments.csv	Payment details
reviews.csv	Customer reviews and ratings
products.csv	Product information
sellers.csv	Seller details
geolocation.csv	Geographic information
category_translation.csv	Category translation mapping